Product List Management System Using Doubly and Circular Linked Lists

```
#include <iostream>
#include <string>
using namespace std;
struct Product
{
   int id;
 string name;
   pflroiacet        ;
   Product
   *next;   Product
}; *prev;
Product  *dHead  =  NULL;
Product *cHead = NULL;

// Insert into Doubly Linked List
void insertDoubly(int id, string name, float price)
{
   Product *n = new Product{id, name, price, NULL, NULL};
   if (dHead == NULL)
   {
      dHead = n;
      return;
   }
 Product *temp = dHead;
while (temp->next != NULL)
temp = temp->next;
 temp->next = n;
n->prev = temp; }
```

```
// Insert into Circular Linked List
void insertCircular(int id, string name, float price)
{
    Product *n = new Product{id, name, price, NULL, NULL};
    if (cHead == NULL)
    {
        cHead = n;
        n->next = cHead;
        return;
    }
 Product *temp = cHead;
while (temp->next != cHead)
temp = temp->next;
 temp->next = n;
    n->next = cHead;
}
// Display Doubly Linked List
void displayDoubly()
{
    Product *temp = dHead;
    cout << "\nDoubly Linked List:\n";
 if (temp == NULL)
    {
        cout << "No products available.\n";
        return;
    }
 while (temp != NULL)
    {
```

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        cout << temp->id << " "
             << temp->name << " "
             << temp->price << endl;
        temp = temp->next;
    }
}
// Display Circular Linked List
void displayCircular()
{
    cout << "\nCircular Linked List:\n";
    if (cHead == NULL)
    {
        cout << "No products available.\n";
        return;
    }
 Product *temp = cHead;
    do
    {
        cout << temp->id << " "
             << temp->name << " "
             << temp->price << endl;
        temp = temp->next;
 } while (temp != cHead);
}
// Search in Doubly Linked List
void searchDoubly(int id)
{
    Product *temp = dHead;
```

```
 while (temp != NULL)
    {
        if (temp->id == id)
        {
            cout << "Product found: " << temp->name
                 << ", Price: " << temp->price << endl;
            return;
        }
        temp = temp->next;
    }
    cout << "Product not found.\n";
}
// Search in Circular Linked List
void searchCircular(int id)
{

    if (cHead == NULL)
    {
        cout << "Product not found.\n";
        return;
    }
 Product *temp = cHead;
    do
    {
        if (temp->id == id)
        {
            cout << "Product found: " << temp->name
                 << ", Price: " << temp->price << endl;
            return;
        }
```

```
        temp = temp->next;
 } while (temp != cHead);
    cout << "Product not found.\n";
}
// Delete from Doubly Linked List
void deleteDoubly(int id)
{
    Product *temp = dHead;
 while (temp != NULL && temp->id != id)
        temp = temp->next;
 if (temp == NULL)
    {
        cout << "Product not found.\n";
        return;
    }
 if (temp->prev != NULL)
        temp->prev->next = temp->next;
    else
         dHead = temp->next;
   if (temp->next != NULL)
        temp->next->prev = temp->prev;
 delete temp;
 cout << "Product deleted from Doubly Linked List.\n";
}
// Delete from Circular Linked List
void deleteCircular(int id)
{
    if (cHead == NULL)
    {
```

```
    cout<<"Productlistis empty.\n";
    return;
}
Product *temp = cHead;
Product *prev = NULL;
do
{
    if (temp->id == id)
    {
        if(temp==cHead&& temp->next == cHead)
        {
            cHead = NULL;
        }
        else
        {
            if (prev == NULL)
            {
                Product*last= cHead;
                while(last->next != cHead)
                    last = last->next;
                cHead = cHead->next;
                last->next = cHead;
            }
            else
            {
                prev->next = temp->next;
            }
        }
```

```
            delete temp;
            cout << "Product deleted from Circular Linked List.\n";
            return;
        }
        prev = temp;
        temp = temp->next;
 } while (temp != cHead);
    cout << "Product not found.\n";
}
int main()
{
    int choice, id;
 string name;
    float price;
    do
    {
        cout << "\n--- Product List Management ---\n";
        cout << "1. Insert Product\n";
        cout << "2. Display Doubly Linked List\n";
        cout << "3. Display Circular Linked List\n";
        cout << "4. Search Product\n";
        cout << "5. Delete Product\n";
        cout << "6. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;
        switch (choice)
        {
        case 1:
            cout << "Enter Product ID: ";
```

```
        cin >> id;
        cout << "Enter Product Name: ";
        cin >> name;
        cout << "Enter Product Price: ";
        cin >> price;
        insertDoubly(id, name, price);
        insertCircular(id, name, price);
        cout << "Product inserted into both lists.\n";
        break;
    case 2:
        displayDoubly();
        break;
    case 3:
        displayCircular();
        break;
    case 4:
        cout << "Enter Product ID to search: ";
        cin >> id;
        cout << "Search in Doubly Linked List:\n";
        searchDoubly(id);
        cout << "Search in Circular Linked List:\n";
        searchCircular(id);
        break;
    case 5:
        cout << "Enter Product ID to delete: ";
        cin >> id;
        deleteDoubly(id);
        deleteCircular(id);
        break;
```

```
        case 6:
            cout << "Exiting program.\n";
            break;
        default:
            cout << "Invalid choice.\n";
        }
    } while (choice != 6);
    return 0;
}
```